Unit 2: Algebraic Expressions

1. Use the expanded form to generate an equivalent expression for $\frac{(9x)(7x^5)}{3x^3}$, where $x \neq 0$.

$$\frac{63x^6}{3x^3} = 21x^3$$

2. Find the product of $(2x^3 - 3b^5)^2$

$$(2x^3 - 3b^5)(2x^3 - 3b^5)$$

$$4x^6 - 6x^3b^5 - 6x^3b^5 + 9b^{10}$$

$$4x^6 - 12x^3b^5 + 9b^{10}$$

3. Use the order of operations to simplify the polynomial and describe each step you perform.

$5\left(\frac{2}{5}x - 2y^3\right) + (3y^2 + 7)$ – Distribute 5

$2x - 10y^3 + 3y^2 + 7$

$-10y^3 + 3y^2 + 2x + 7$ – Rewrite in standard form

4. Find the quotient if $x \neq 0$ and $y \neq 0$

$$\frac{75x^2y^2 - 25x^4 2y^2 - 50x^4\,y^8}{-5x^2y^4}$$

$\frac{75x^2y^2}{-5x^2y^4}$ $\frac{-25x^4 2y^2}{-5x^2y^4}$ $\frac{-50x^4y^8}{-5x^2y^4}$

$-15y^{-2}$ $10x^2y^{-2}$ $10x^2y^4$

$$\frac{-15}{y^2} + \frac{10x^2}{y^2} + 10x^2y^4$$